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| <b>LESSON PLAN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>COURSE: MDM4U</b>                                                       |
| <b>Date:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Unit of Study: Unit 1</b>                                               |
| <b>Materials/Resources:</b><br><b>Mathmatics of data management study guide</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Lesson Focus/Theme:</b><br><b>Chapter 2: Statistics of One variable</b> |
| <b>Prior Knowledge: Database</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                            |
| <b>Overall Expectations:</b><br><br>By the end of this lesson, students will:<br><br>Analyse, interpret, and draw conclusions from one-variable data using numerical and graphical summaries                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                            |
| <b>Specific Expectations:</b><br><br>1.1 recognize that the analysis of one-variable data involves the frequencies associated with one attribute, and determine, using technology, the relevant numerical summaries (i.e., mean, median, mode, range, interquartile range, variance, and standard deviation)<br><br>1.2 determine the positions of individual data points within a one-variable data set using quartiles, percentiles, and z-scores, use the normal distribution to model suitable one-variable data sets, and recognize these processes as strategies for one-variable data analysis<br><br>1.3 generate, using technology, the relevant graphical summaries of one-variable data (e.g., circle graphs, bar graphs, histograms, stem-and-leaf plots, boxplots) based on the type of data provided (e.g., categorical, ordinal, quantitative) |                                                                            |

**Learning Goals:**

By the end of this lesson, students will know how to calculate the numeric summaries, position of individual data points and graphical summaries

**Success Criteria:**

By the end of this lesson, I will be able to know how to calculate the numeric summaries

By the end of this lesson, I will be able to position the individual data points

By the end of this lesson, I will be able to know the graphical summaries

**Assessment Strategies:** Tests, quiz and project

**Time:**  
**10 min**

**Introduction/Minds-On:**

Answer questions students don't understand from last class if any

**Time:**  
**100 min**

**Lesson:**

Chapter 2: Statistics of one variable  
2.1 Data analysis with graphs

2.2 Indices

2.3 Sampling techniques

**Time:**

10 min

**Extension / Wrap-up:**

Review the key concepts and formula for this lesson

**Notes / Reminders / Homework:**

**Lesson Reflection:**